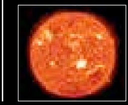
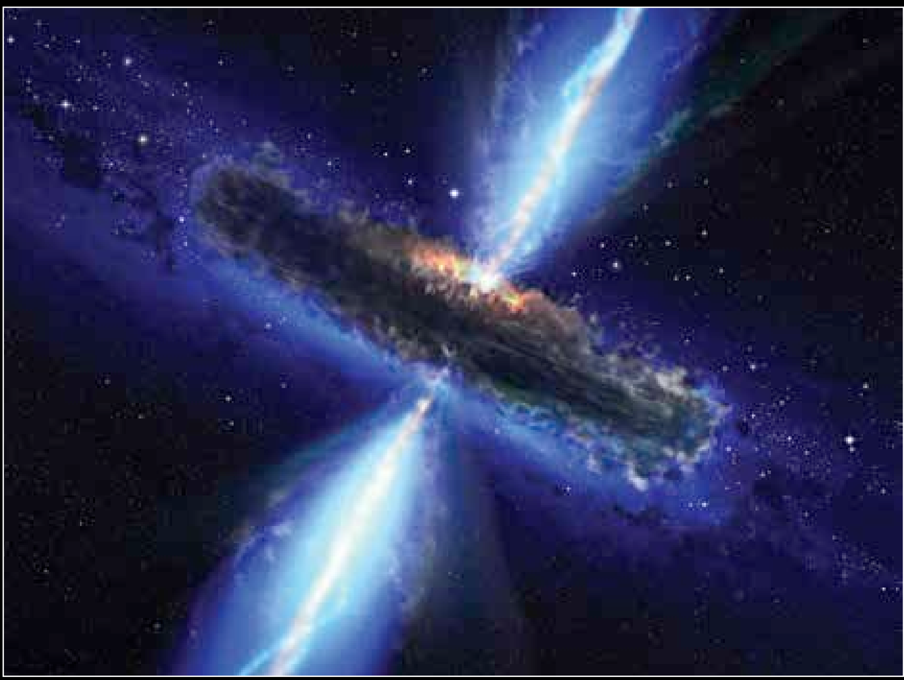


Sun Bombed!
You're never too big to get photo bombed. Watch how the moon photobombs the sun **http://dgit.in/sphbmb**

First look inside an asteroid
By studying a couple of images taken between 2001 and 2013, scientists have been able to peek inside an asteroid for the first time **http://dgit.in/InAnAsteroid**

Through the Universe, Darkly

Matter comes in many forms but nothing is more exciting than the exotic kind. We explain why exotic matter has gripped the scientific imagination despite its theoretical strangeness



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"The finder of a new elementary particle used to be rewarded by a Nobel Prize, but such a discovery now ought to be punished by a 10,000 dollar fine."
- Willis Lamb, Nobel Prize Lecture 1955

In 1928, British physicist Paul Dirac shook the foundations of modern physics and theorised the existence of sub-atomic particles that had never before been imagined. The elegance of his mathematical brilliance posited the existence of various types of matter without the support of any physical evidence. Relying solely on logic, observation and mathematical deduction, Dirac crafted equations that described how these matter particles would behave in our universe.

In 1932, an American physicist - Carl Anderson - stumbled across the physical reality that Dirac had prophesied. While running a study on cosmic rays, Anderson discovered that the shower of subatomic particles colliding with the Earth's atmosphere exhibited strange properties. In his photographs, Anderson observed a particle that had the same mass as an electron, but was deviated against the magnetic field. He had stumbled on one of Dirac's predicted particles – a positron.

Paul Dirac was awarded a Nobel Prize in 1933 for his formulation of the Dirac equation, which predicted the existence of positrons and other Fermion (named after Enrico Fermi) particles, giving birth to the idea of antimatter. As one of the early explorers of quantum theory, Dirac pushed forward the quest to broaden physics beyond any known threshold. Since then hundreds of scientists have attempted to conceptualise, discover and understand different types of matter – subatomic, dark and exotic.

What's the matter anyway?

Before we dive deeper into the esoteric world of scientific jargon, let's first understand what "matter" is.

Scientists have a slightly demented relationship with the universe. Physicists, in particular, spend decades reconciling their awestruck curiosity of the universe with their egotistical need to decipher it. The most fundamental of all is matter – that ubiquitous substance from which the universe takes its form. And even with something as basic and self-evident as matter, these experts still complicate reality with theoretical ideas.

For the longest time in human history, matter was thought to comprise the four natural elements: fire, water, earth and air (and ether, if you consider Greek theorists). However, the modern age moved beyond such archaic thinking and realised that matter is perpetually indivisible, with the atom being recognised as the fundamental unit of matter.

And for a while, things were good. But not for long.